

A Machine Learning Approach to Predicting No-Shows in a Philippine Pediatric Therapy Clinic

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Abstract—This study addresses appointment non-attendance at a private pediatric therapy clinic in Isabela, Philippines, analyzing 1,506 records to identify drivers of patient no-shows. Using a retrospective approach, the research examined the influence of financial commitment, temporal patterns, environmental disruptions, and historical behavior. Results identify typhoon signal levels as the most significant predictor of non-attendance, with no-show rates surging from 4.87% during clear weather to 40.77% during active tropical cyclones. Statistical analysis also revealed a strong "recency effect," where the outcome of the immediately preceding appointment significantly influenced current attendance. Conversely, patient demographics, clinical diagnoses, and session-specific down payments failed to show statistically significant associations with adherence. Three machine learning algorithm: Logistic Regression, Random Forest, and Gradient Boosting, were evaluated for predictive performance. While Logistic Regression maximized risk detection (Recall: 0.708), Gradient Boosting emerged as the optimal operational model, achieving the highest F1-Score (0.565) and providing a balanced equilibrium between precision and recall. These findings provide a framework for clinics to implement risk-stratified scheduling and weather-contingent operational protocols in the Philippine context

Index Terms—Appointment no-shows, binary classification, machine learning, pediatric therapy, predictive modeling, healthcare analytics

I. INTRODUCTION

Appointment non-attendance represents a persistent challenge in healthcare delivery systems worldwide, with significant implications for operational efficiency, revenue sustainability, and patient care quality. In pediatric therapy settings, where consistent attendance is crucial for developmental progress and treatment effectiveness, the problem of no-shows becomes particularly acute. Missed appointments generate cascading effects throughout the healthcare ecosystem: therapists experience fragmented schedules with unused time slots that could have been allocated to other patients, clinics face revenue losses from unpaid services that cannot be recovered, and patients themselves miss critical intervention opportunities during sensitive developmental windows. While this challenge exists across healthcare systems globally, it manifests with unique characteristics in the Philippine context, where factors such as tropical weather disruptions, cultural holiday patterns, and economic constraints create a distinct attendance behavior profile that remains underexplored in existing literature.

This study addresses the critical gap in understanding appointment adherence behavior specific to Philippine pediatric therapy clinic by developing a machine learning-based prediction model for no-shows. The research transforms routine administrative data, typically collected but underutilized, into actionable systems that can guide clinic management decisions. By systematically analyzing

appointment schedules, payment behaviors, patient demographics, and external environmental factors, the study seeks to identify which variables most strongly predict non-attendance and how these variables interact to create high-risk appointment scenarios.

Specifically, the study aimed to achieve the following objectives:

1. To determine the effect of financial commitment variables such as down payments and updated payment status on the likelihood of appointment adherence.
2. To examine the statistical influence of environmental and temporal factors, including government-declared holidays, typhoon signal warnings, and time-of-day scheduling, on patient attendance rates.
3. To identify demographic risk patterns by analyzing whether factors such as age, clinical diagnosis, or geographic distance from the clinic are significantly associated with cancellation frequencies.
4. To quantify and rank predictor importance using feature importance analysis to determine which categories of variables (financial, environmental, or demographic) exert the strongest influence on patient attendance outcomes.

To operationalize these objectives and provide a structured analytical direction, the study is further guided by the following research questions:

1. To what extent does the status of financial commitment, specifically the settlement of down payments, influence the probability of appointment adherence?
2. Are there significant temporal variations in non-attendance rates based on the day of the week or specific time-of-day blocks?
3. How do external environmental determinants, specifically government-declared holidays and PAGASA typhoon signal warnings, impact the statistical likelihood of patient cancellations?
4. Do significant differences in cancellation rates exist across different patient profiles defined by age groups, clinical diagnoses, and residential distance from the clinic?
5. Does a patient's historical attendance record, specifically the frequency of prior cancellations, serve as a significant predictor for future appointments non-attendance?

II. LITERATURE REVIEW

The application of machine learning to predict healthcare no-shows is driven by the need to improve operational efficiency and patient care. Dunstan et al. [1] analyzed nearly 396,000 pediatric appointments in Chile using gradient boosting and random forest algorithms, achieving prediction

accuracies exceeding 75%. Their findings identified historical cancellation patterns and distance from the clinic as primary predictors. However, the study focused on a public hospital setting, omitting financial commitment variables relevant to private practice and failing to account for environmental factors such as tropical weather or cultural holiday patterns specific to the Philippines. Building on this, Hathaway et al. [2] applied XGBoost models to pediatric data spanning pre-pandemic and pandemic periods, demonstrating that appointment characteristics, specifically time of day and day of week, held strong predictive power. While validating the importance of temporal features, their research was conducted in a non-Philippine setting and does not capture local temporal dynamics, such as typhoon seasons or specific national holidays.

Incorporating environmental factors, Liu et al. [3] integrated local weather data into deep learning models, finding that adverse weather conditions could increase no-show probabilities by up to 40%. This establishes a precedent for using environmental variables in prediction models, though the U.S.-based weather patterns examined differ substantially from the Philippine tropical climate. The current research extends this work by integrating typhoon signal warnings and rainy season indicators alongside government-declared holidays. Similarly, Osorio et al. [4] demonstrated that a patient's past appointment history, specifically the frequency and recency of previous cancellations, significantly improved model performance, raising AUC scores from 0.65 to over 0.80.

Recent studies have further expanded the scope of no-show prediction. AlSerkal et al. [5] implemented an AI-driven system in UAE primary health centers, achieving a 50.7% reduction in no-show rates by categorizing appointments based on risk. This highlights the practical utility of predictive models in clinical operations, yet the study's setting differs significantly from the resource-constrained environment of many Philippine clinics. Similarly, research by Tuan et al. [6] utilized machine learning to identify that lead time, the duration between scheduling and the appointment, was the most influential factor in missed appointments, suggesting that shorter scheduling windows could improve adherence. This finding is particularly relevant for the current study, which examines lead time as a key operational variable.

Furthermore, studies by Malloy et al. [7] and Chen [8] have explored the impact of specific clinical departments and patient demographics on no-show rates. Malloy et al. found that no-show rates varied significantly by department, with higher rates in neurology and lower rates in surgery, while Chen emphasized the role of socioeconomic factors, noting that patients with lower income and those relying on public insurance were more likely to miss appointments. These findings underscore the importance of including clinical diagnosis and payment status as predictors in the current model.

The financial implications of missed appointments have been well-documented. Gier [9] estimated that no-shows cost the U.S. healthcare system \$150 billion annually, while a separate study by Kheirkhah et al. [10] highlighted the operational inefficiencies caused by unused resources. These economic factors are crucial for private clinics in the Philippines, where revenue loss directly impacts sustainability. Collectively, these studies provide a

methodological framework but overlook specific Philippine healthcare dynamics. Existing literature largely ignores the interaction between tropical weather systems, culturally specific holidays, and financial mechanisms prevalent in private therapy clinics. This study fills these gaps by developing a predictive model tailored to this context, integrating a comprehensive feature set that spans financial, environmental, demographic, and behavioral dimensions to identify high-risk appointment profiles.

III. METHODOLOGY

A. Participants

The study utilized data obtained from a private pediatric therapy clinic located in the province of Isabela, Region II (Cagayan Valley), Philippines. The "participants" in this retrospective analysis consisted of the pediatric patient population represented within the clinic's historical operational records. Prior to the commencement of data collection, formal informed consent was secured from the clinic's administration to access, extract, and analyze appointment logs and client demographic profiles. To uphold ethical standards and patient confidentiality, all personally identifiable information (PII) was anonymized at the source. Strict data privacy protocols were enforced to ensure that no sensitive personal data was exposed or compromised during the research process.

B. Data Collection

Data collection was conducted retrospectively using administrative records from a private pediatric therapy clinic in Isabela, Philippines. The dataset spans from February to October 2025, capturing daily appointment activities across diverse seasonal contexts, including the transition from the dry season to the wet season and periods affected by typhoons and national holidays.

Primary internal data were aggregated from the clinic's digital therapy schedules and client information sheets. These records were systematically logged and managed using Google Sheets, which served as the central repository for structured data entry and storage. This internal dataset was then supplemented with secondary external data: historical weather information was sourced from Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) bulletins to track typhoon signals, while holiday declarations were verified against official government gazettes.

The dataset comprises a comprehensive set of features categorized into five distinct domains: demographics, appointment details, booking behavior, financial information, and external environmental factors. Table 1 provides a detailed summary of the specific variables extracted and utilized in the analysis.

TABLE 1. SUMMARY OF EXTRACTED VARIABLES

Variable	Description
id	Unique identifier for each patient
age	Patient's age in years at time of appointment
age_group	Categorical grouping of age
gender	Biological sex of the patient
diagnosis	Medical condition or reason for therapy/consultation

Variable	Description
distance_km	Calculated distance (in km) from patient's residence to clinic
distance_category	Categorical grouping of distance
date	Scheduled appointment date
appointment_start_time_24h	Start time in 24-hour format
appointment_end_time_24h	End time in 24-hour format
duration_minutes	Length of session in minutes
service_type	Type of therapy provided (OT or ST)
appointment_hour	Integer hour of appointment
time_of_day	Time category (Morning, Lunch, Afternoon)
day_of_the_week	Named day of appointment
is_weekend	Binary flag (1 = Weekend, 0 = Weekday)
attendance_status	Raw attendance label from system
lead_time_days	Days between booking date and appointment date
lead_time_is_missing	Binary flag if booking date unavailable
reminder_sent	Binary flag indicating reminder was sent
reminder_date	Date reminder was sent
reminder_time_lag	Days before appointment when reminder was sent
has_down_payment	Binary flag (1 = Deposit made, 0 = None)
down_payment_amount	Deposit amount in PHP
down_payment_date	Date deposit was recorded
payment_status	Payment status (Arrears, Updated)
payment_method	Payment channel (Cash, GCash, BDO)
is_typhoon_risk	Binary flag indicating typhoon presence
typhoon_signal_level	PAGASA signal level (0 = None, 1–5 = Severity)
is_rainy_season	Binary flag for wet season months
is_holiday	Binary flag indicating public holiday
holiday_type	Classification of holiday

C. Operational Definitions

To ensure the reproducibility and clarity of the analysis, the following operational definitions were established for the key variables used in the study:

1. Target Variable

No-Show (*is_noshow*): A binary classification where 1 denotes that the patient failed to attend the scheduled therapy session without a cancellation notice, and 0 denotes that the patient attended the session ("Show").

2. Demographic Variables

- Age: The chronological age of the patient in years at the time of the appointment, calculated from their date of birth.
- Distance (*distance_km*): The geodesic distance in kilometers between the patient's registered residential address and the clinic location, calculated using geospatial mapping coordinates.
- Diagnosis Category: The classification of the patient's primary medical condition as recorded in their intake forms.

3. Appointment & Temporal Variables

- Lead Time (*lead_time_days*): The integer count of days elapsed between the date the appointment was scheduled (booking date) and the actual date of the appointment.
- Time of Day: A categorical classification of the appointment start time, operationalized as Morning (8:00 AM - 11:59 AM), Lunch (12:00 PM - 1:00 PM), and Afternoon (1:00 PM - 6:00 PM).
- Weekend Indicator: A binary variable where 1 indicates the appointment falls on a Saturday or Sunday, and 0 indicates a weekday (Monday-Friday).

4. Financial Variables

- Down Payment Status (*has_down_payment*): A binary indicator where 1 signifies that a partial monetary deposit was recorded for the specific appointment prior to the session, and 0 signifies no deposit was made.
- Payment Status: A categorical label describing the patient's account standing, operationalized as Updated (no outstanding balance) or Arrears (has outstanding balance).

5. Environmental Variables

- Typhoon Signal (*typhoon_signal_level*): The highest Tropical Cyclone Wind Signal (TCWS) level (scale 0-5) active in the Isabela region on the appointment date, as reported by PAGASA bulletins.
- Holiday (*is_holiday*): A binary variable where 1 indicates the appointment date coincides with an official government-declared Regular Holiday or Special Non-Working Holiday, and 0 indicates a regular working day.

6. Historical Behavior Variables

- Prior No-Show Count: The cumulative integer count of appointments missed by the specific patient prior to the current scheduled session.
- No-Show Ratio: The calculated proportion of a patient's historical missed appointments divided by their total number of scheduled appointments up to that date.

D. Data Cleaning

To ensure data integrity, analytical consistency, and suitability for machine learning modeling, a systematic data cleaning and feature engineering process was conducted. The procedures implemented are described below.

Handling Missing Values

Missing Lead Time. A binary indicator variable, *lead_time_is_missing*, was created to flag cases where *lead_time_days* was null. Subsequently, missing *lead_time_days* values were imputed with 0. This approach preserves structural information, as missing lead time often corresponds to appointments without formal booking records or down payments, representing a potentially meaningful behavioral pattern.

Missing Payment Method. Missing values in *payment_method* were replaced with the category "None." This imputation reflects the absence of financial commitment, typically indicating that no down payment was made.

Target Variable Creation. A binary target variable, *is_noshow*, was generated from the original *attendance_status* field. Appointments marked as "Yes" (no-show) were encoded as 1, while attended appointments ("No") were encoded as 0. This transformation standardized the outcome variable for binary classification modeling.

Feature Engineering

Categorical Variables. Several categorical features were derived to enhance interpretability and predictive capacity:

Weekend Flag. A binary variable, *is_weekend*, was created to identify appointments scheduled on Saturdays or Sundays.

Time-of-Day Categorization. The integer *appointment_hour* was extracted from the appointment start time. Based on this, a categorical variable *time_of_day* was

created with the following bins: Morning (before 12:00); Lunch (exactly 12:00); Afternoon (after 12:00).

Distance Categories. The continuous variable `distance_km` was discretized into `distance_category` bins: Near (0-10 km); Moderate (10-30 km); Far (>30 km)

Age Groups. The continuous variable `age` was categorized into `age_group` bins: 3-4 years; 5-6 years; 7-9 years.

Historical Behavior Features. To capture longitudinal attendance behavior, chronological feature engineering was performed. The `date` variable was first converted to `datetime` format, and the dataset was sorted by `patient_id` and `appointment_date`. The following historical features were generated:

previous_appointment_count: Cumulative number of prior appointments per patient.

last_appt_noshow: Binary lag feature indicating whether the immediately preceding appointment was a no-show.

previous_noshow_count: Running total of prior no-shows (excluding the current appointment).

noshow_ratio: Ratio of previous no-shows to previous appointments (set to 0 for patients with no prior history).

Log Transformations

To address high positive skewness in selected numerical variables, logarithmic transformation using `log1p` was applied to: `duration_minutes`, `previous_appointment_count`, `lead_time_days`, and `distance_km`.

The transformation compresses extreme values and reduces the influence of outliers. Notably, `duration_minutes` exhibited substantial skewness (10.43), warranting normalization to stabilize variance and improve model performance.

E. Statistical Analysis

To examine the relationships between predictor variables and appointment non-attendance, a structured statistical analysis framework was employed prior to machine learning modeling.

Pearson correlation analysis was conducted to measure the linear association between numerical predictor variables and the binary target variable (`is_noshow`). This procedure was used to evaluate the direction and strength of relationships between continuous features and appointment non-attendance. Correlation coefficients were interpreted based on magnitude and direction to identify potential predictive relevance.

For categorical variables, Chi-Square (χ^2) tests of independence were performed to assess whether significant associations existed between categorical predictors and attendance outcomes.

All hypothesis testing was conducted at a significance level of $\alpha = 0.05$. Variables with p-values below this threshold were considered statistically associated with appointment non-attendance. To formally evaluate the

study's research questions, multiple inferential statistical tests were applied: Independent Samples t-tests were used to compare mean no-show rates between two groups. One-Way ANOVA (Analysis of Variance) was conducted to examine differences in cancellation rates across multiple categories. These tests allowed for systematic evaluation of whether statistically significant differences existed across financial, temporal, environmental, demographic, and historical dimensions.

To complement inferential testing and enhance interpretability, exploratory data visualizations were generated, including: (1) Bar charts to compare cancellation rates across categorical groups and; (2) Line plots to visualize temporal trends. Visual analytics supported statistical findings by providing intuitive representations of patterns and group differences.

IV. RESULTS

This section presents the findings from the analysis of 1,506 pediatric therapy appointments at a Philippine clinic. All analyses were conducted using Python with a significance level of $\alpha = 0.05$.

Dataset Overview

The dataset comprises 1,506 appointment records collected from February to October 2025. After data cleaning and feature engineering, the final dataset contains 40 variables across five categories: patient demographics, appointment details, financial information, environmental factors, and historical behavior metrics.

TABLE 2. DATASET COMPOSITION

Characteristic	Value
Total appointments	1,506
Unique patients	73
Date range	February 24 - October 15, 2025
Total features	40 (26 original + 14 engineered)
No-show rate	7.97% (120 no-shows)

The dataset is imbalanced with 92.03% attendance (1,386 showed up) versus 7.97% no-shows (120 missed appointments).

Descriptive Statistics

As shown in Table 3, sessions were highly standardized in duration, with a median of 60 minutes and minimal variation ($SD = 5.60$), although a small number extended up to 120 minutes. Down payment amounts showed substantial variability (Mean = ₱860.71, $SD = ₱675.16$), with many appointments having no deposit (25th percentile = ₱0), indicating inconsistent financial commitment across patients. Typhoon exposure was generally low (Mean = 0.17), with most appointments occurring under Signal 0 conditions, though signal levels reached up to 2 in some cases. Lead time averaged 6.11 days, suggesting that most bookings were made within one week of the appointment, while reminders were typically sent approximately 2-3 days prior.

TABLE 3. SUMMARY OF NUMERICAL VARIABLES

Variable	Mean	Std	Min	25%	50% (Median)	75%	Max
duration_minutes	60.54	5.60	60.00	60.00	60.00	60.00	120.00
down_payment_amt	860.71	675.16	0.00	0.00	1,000.00	1,500.00	3,500.00
typhoon_signal	0.17	0.54	0.00	0.00	0.00	0.00	2.00
lead_time_days	6.11	3.88	0.00	3.00	5.00	8.00	30.00
reminder_lag	2.50	0.93	0.00	2.00	2.00	3.00	13.00
age (Years)	6.06	1.47	3.00	5.00	6.00	7.00	9.00
distance_km	22.81	19.79	2.73	2.73	17.13	31.73	59.73

Patients were primarily young children (Median age = 6 years) with limited dispersion (3-9 years range). Geographic distance to the clinic varied widely (Mean = 22.81 km, SD = 19.79), ranging from nearby residences (~3 km) to patients traveling nearly 60 km, indicating substantial variability in travel burden.

TABLE 4. SUMMARY OF CATEGORICAL VARIABLES

Variable	Categories (Top/Distribution)	Unique Values
date	83 appointment dates (range: 6-40 appointments per date)	83
appointment_start_time_24h	Most common: 11:00 (223), 10:00 (212), 13:00 (212), 14:00 (201)	13
appointment_end_time_24h	Most common: 12:00 (221), 14:00 (213), 11:00 (212), 15:00 (200)	14
day_of_the_week	Sun (467), Sat (408), Tue (221), Mon (217), Wed (182), Fri (11)	6
service_type	OT (806), ST (700)	2
has_down_payment	Yes (1033), No (473)	2
down_payment_date	193 unique transaction dates	193
payment_status	Arrears (1068), Updated (438)	2
payment_method	GCash (816), BDO (185), Cash (26)	3
is_holiday	No (1420), Yes (86)	2
holiday_type	Ordinary Day (1420), Observance (36), Special Non-working (24), Regular (17), Special Working (9)	5
is_typhoon_risk	No (1376), Yes (130)	2
is_rainy_season	Yes (878), No (628)	2
attendance_status	Attended (1386), No-show (120)	2
reminder_sent	Yes (1506)	1
reminder_date	76 unique reminder dates	76
gender	Male (1176), Female (330)	2
diagnosis	Most common: SLD (276), ASD (272), Intellectual Disability (139), ADHD (132)	12

Appointments were predominantly scheduled on weekends, with 467 sessions on Sundays and 408 on Saturdays (58.1% of all sessions), compared to substantially fewer weekday appointments (e.g., 221 on Tuesdays and only 11 on Fridays). This concentration suggests potential temporal clustering effects in attendance behavior. Service distribution was relatively balanced (806 OT; 700 ST). The clinic operates primarily with 60-minute sessions (98.9%).

Most appointments involved a down payment (1,033 vs. 473 without); however, 1,068 cases were classified as “Arrears,” indicating that initial deposits did not necessarily reflect fully updated payment status. GCash was the dominant payment method (816 transactions), accounting for nearly 80% of transactions, suggesting strong reliance on digital payments.

Environmental conditions exhibited limited variation across the dataset. A total of 1,420 appointments (94.3%) occurred on ordinary, non-holiday dates, while only 86 were scheduled on holidays. Similarly, 1,376 appointments

(91.4%) took place during periods without typhoon activity, with 130 conducted under typhoon-risk conditions; signal levels ranged from 0 to 3, with Signal 2 being the most frequent when present. Additionally, 878 appointments (58.3%) occurred during the rainy season (June-November).

Attendance outcomes were imbalanced, with 1,386 attended appointments compared to 120 no-shows (8%), indicating a minority-class classification scenario for predictive modeling. The patient population was predominantly male (1,176 vs. 330 female) and clinically diverse, with SLD (276) and ASD (272) as the most common diagnoses.

Exploratory Data Analysis

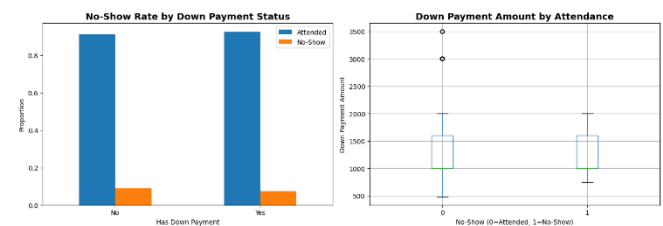


Fig. 1. Analysis of Financial Commitment and Attendance

Visual inspection of the financial data indicates that the mere presence of a standard down payment does not immediately differentiate between attendees and non-attendees. The proportion of no-shows remains consistent at approximately 8-10% across both groups, those who paid a deposit and those who did not. Furthermore, the median down payment amount is identical (PHP 1,000) for both outcomes, suggesting that the standard deposit magnitude commonly used by the clinic may not be distinct enough to visually separate the two groups. However, a notable pattern emerges at higher payment thresholds: while rare, every recorded instance where the IE session fee (PHP 3,000) was paid in advance was associated with 100% attendance. This observation suggests that while partial deposits show little variance in outcome, full financial commitment aligns with perfect appointment adherence in this dataset.

As shown in Figure 2, analysis of temporal patterns reveals a distinct inverse relationship between appointment volume and non-attendance. Sundays and Saturdays record the highest patient volumes (467 and 408, respectively) yet maintain the lowest no-show rates (4-6%). Conversely, mid-week appointments exhibit significant friction; Wednesday, despite having less than half the volume of Sunday, registers the highest no-show rate at 17%. Aggregated data confirms that patients are approximately twice as likely to miss a

weekday appointment compared to a weekend session. Intraday analysis further indicates a linear improvement in attendance as the day progresses. The no-show rate is highest during early morning slots, peaking at 25%, nearly triple the daily average, before steadily declining to 0% by 5:00 PM.

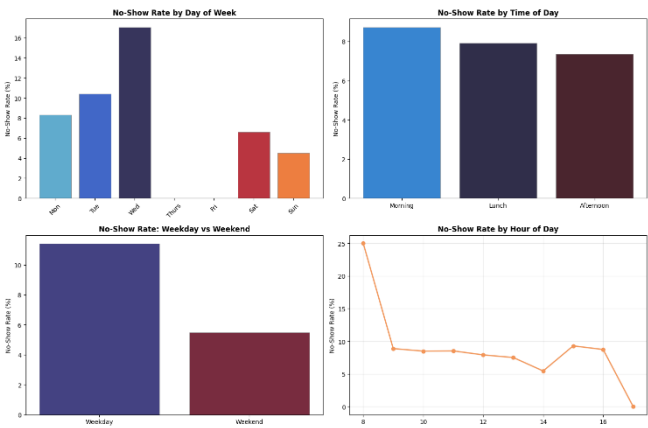


Fig. 2. Temporal Analysis of Attendance

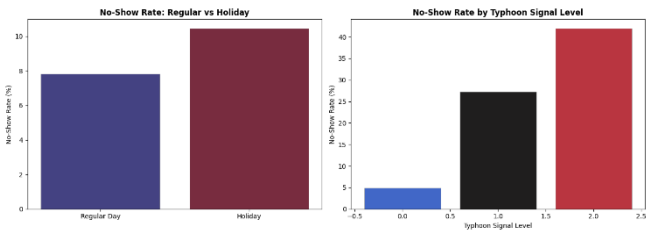


Fig. 3. Environmental Impact on Attendance

Examination of external environmental variables confirms that non-controllable disruptions significantly impact patient attendance. As seen in Figure 3, appointments scheduled on dates declared as public holidays exhibit notably elevated no-show rates compared to regular working days, likely due to family travel or conflicting social engagements. Similarly, a direct positive correlation was observed between tropical cyclone warnings and cancellation rates. As the PAGASA Typhoon Signal level increases, the probability of non-attendance rises sharply, validating the hypothesis that adverse weather conditions act as a primary physical barrier to clinic access.

As seen in Figure 4, The analysis of patient demographics and clinical profiles uncovered specific high-risk segments. In terms of diagnosis, families managing Global Developmental Delay (GDD) or Oppositional Defiant Disorder (ODD) exhibited the highest non-attendance frequencies, peaking between 12% and 13%. Age-stratified analysis further indicated that older children (ages 7-9) are significantly more likely to miss appointments (nearly 10%) compared to the younger 3-4-year-old cohort (~7%). Geographically, a notable pattern emerged regarding travel distance: patients living a "Moderate" distance from the clinic had a higher no-show rate (~8.5%) than those classified as living "Far" (~7.5%). This observation suggests that families traveling long distances may prioritize rigorous planning,

whereas those at intermediate distances might be more susceptible to immediate disruptions or miscalculations in travel time.

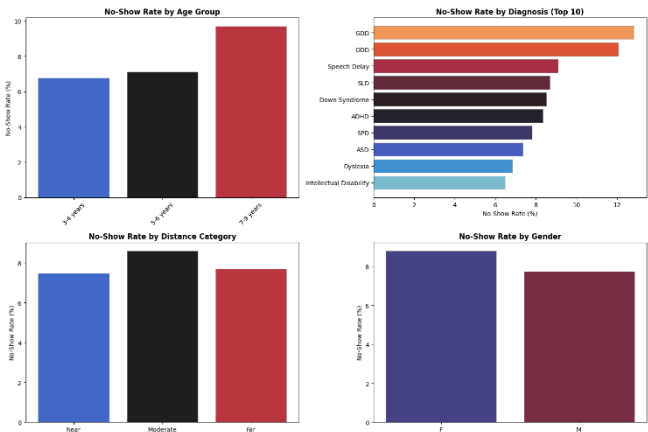


Fig. 4. Demographic and Clinical Risk Factors

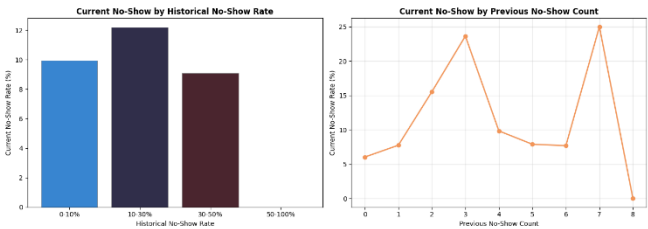


Fig. 5. Historical Attendance Patterns

Analysis of historical attendance patterns highlights the predictive value of past behavior. Patients maintaining a historical no-show rate between 10% and 30% emerged as the highest-risk cohort, exhibiting a current non-attendance rate of 12%. A distinct pattern was observed among serial absentees: those with exactly seven prior missed appointments displayed a sharp increase in non-attendance, reaching 25%. However, beyond this threshold (eight or more misses), the recorded no-show rate drops to 0%. This abrupt decline is likely an artifact of administrative intervention, such as the clinic terminating services for non-compliant clients, rather than a spontaneous behavioral correction.

Distribution and Correlation Analysis

Skewness analysis revealed significant positive skews in several continuous variables. Duration (skewness = 10.43) showed extreme right skew due to a small number of 120-minute double sessions. Previous appointment count (1.52), lead time days (1.45), and distance (0.64) also exhibited positive skew. Log transformation ($\log_{1+x}$) was applied to these variables to normalize distributions for modeling. The histograms display right-skewed distributions for previous_appointment_count, lead_time_days, and distance_km, as shown in Figure 5.

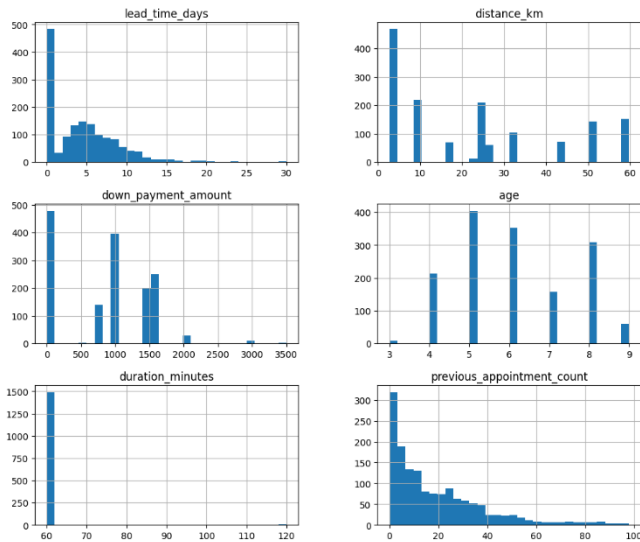


Fig. 6. Distribution and Skewness of Numerical Features

TABLE 5. TOP NUMERICAL FEATURES WITH THE STRONGEST CORRELATION

Rank	Feature	Correlation (r)
1	typhoon_signal_level	0.375
2	last_appt_noshow	0.141
3	previous_noshow_count	0.067
4	log_previous_appointment_count	0.064
5	previous_appointment_count	0.063
6	reminder_time_lag	0.049
7	age	0.047
8	lead_time_days	0.036
9	noshow_ratio	0.034

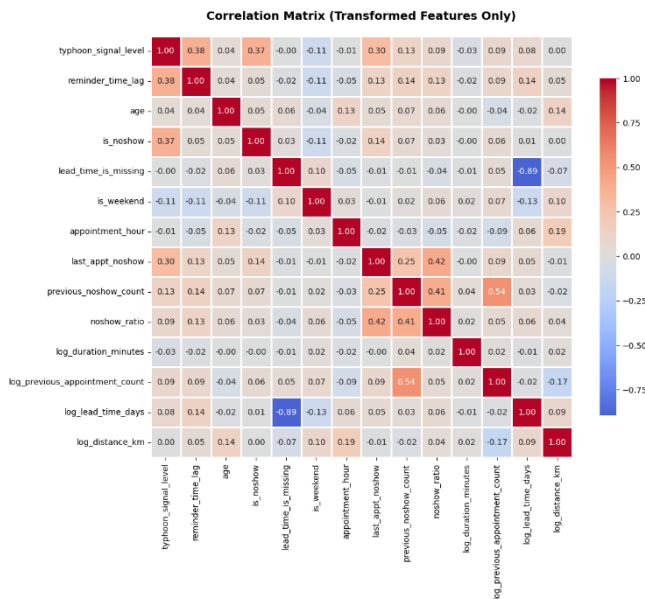


Fig. 7. Pearson Correlation Heatmap of Predictor Variables

The correlation analysis, as shown in Table 5 and Figure 7, identifies external environmental factors and recent patient behavior as the primary linear drivers of non-attendance. Specifically, typhoon_signal_level exhibits the strongest positive correlation ($r=0.375$), indicating that severe weather acts as the most significant single barrier to attendance. A notable "recency effect" also emerged, where the outcome of the immediately preceding appointment (last_appt_noshow, $r=0.141$) proved to be a stronger predictor than a patient's

long-term aggregate history (previous_noshow_count, $r=0.067$). In contrast, demographic and operational variables, such as patient age, lead time, and reminder lag, showed negligible linear correlations ($r < 0.05$). This lack of strong linear association suggests that these factors are likely to influence attendance through more complex, non-linear interactions.

To determine the statistical significance of the relationships between categorical predictors and patient attendance, Chi-Square (χ^2) tests of independence were conducted. The results, summarized in Table 6, isolate which categorical factors genuinely influence the probability of a no-show.

TABLE 6. TOP CATEGORICAL FEATURES WITH THE STRONGEST CORRELATION

Variable	Chi-Square (χ^2)	p-value
day_of_the_week	31.86	< 0.001
is_weekend	16.75	< 0.001
payment_status	13.29	< 0.001
service_type	0.81	0.368
has_down_payment	0.61	0.435
is_holiday	0.46	0.499
gender	0.26	0.612
time_of_day	0.84	0.657
diagnosis	7.46	0.760

Temporal variables, specifically day_of_the_week and is_weekend, demonstrated highly significant associations with no-show outcomes ($p < 0.001$), confirming that appointment timing is a primary determinant of attendance. In contrast, while payment_status was statistically significant ($p < 0.001$), the presence of a session-specific down payment (has_down_payment) was not ($p = 0.435$), suggesting that overall account standing is more predictive than transactional deposits. Furthermore, variables such as diagnosis ($p = 0.760$) and is_holiday ($p = 0.499$) did not exhibit significant associations, indicating that observed variations in exploratory analysis may be attributable to random sample fluctuations rather than systematic effects.

Statistical Hypothesis Testing

Five hypotheses were tested to identify significant predictors of no-show behavior. Tests included chi-square tests for categorical variables, independent t-tests for binary comparisons, and one-way ANOVA for multi-group comparisons. Results are summarized below.

Research Question 1: To what extent does the status of financial commitment, specifically the settlement of down payments influence the probability of appointment adherence?

H_0 : There is no significant difference in cancellation rates between patients who pay a down payment and those who do not.

H_a : Patients who pay a down payment have significantly lower cancellation rates than those who do not.

To evaluate the impact of financial commitment on patient adherence, the study compared cancellation rates between appointments secured with a down payment and those without. Although patients who paid a deposit exhibited a slightly lower no-show rate (7.55%) compared to those who did not (8.88%), this observed difference of 1.33% was not statistically significant. Both the Chi-Square test ($p=0.435$) and the T-test ($p=0.377$) yielded p-values well above the 0.05 significance threshold. Consequently, the study fails to reject the null hypothesis, leading to the conclusion that the current down payment mechanism does not serve as a statistically significant deterrent against appointment cancellations.

TABLE 7. HYPOTHESIS TESTING RESULT OF FINANCIAL COMMITMENT

Metric	Value
No-Show Rate (With Down Payment)	7.55%
No-Show Rate (No Down Payment)	8.88%
Absolute Difference	1.33%
Chi-Square Statistic	0.6104
Chi-Square p-value	0.4346
T-Test Statistic	0.8834
T-Test p-value	0.3772
Decision ($\alpha = 0.05$)	Fail to Reject H_0

Research Question 2: Are there significant temporal variations in non-attendance rates based on the day of the week or specific time-of-day blocks?

H_0 : Cancellation rates do not differ by day of the week or appointment time.

H_a : Patients who pay a down payment have significantly lower cancellation rates than those who do not.

The statistical analysis leads to the rejection of the null hypothesis, as temporal factors were found to significantly influence attendance, though this effect is driven exclusively by the day of the week rather than the time of day. The ANOVA results for the day of the week were highly significant ($F=6.48$, $p < 0.001$), confirming that the specific day chosen for therapy is a critical predictor of adherence. In contrast, the time of day yielded no statistically significant variance ($p=0.657$). Consequently, the alternative hypothesis is partially supported: while the specific day (e.g., Weekday vs. Weekend) exerts a strong influence on patient behavior, the specific hour block (Morning vs. Afternoon) does not.

TABLE 8. HYPOTHESIS TESTING RESULT OF TEMPORAL PATTERNS

Metric	Value
Highest Day Rate (Wednesday)	17.03%
Lowest Day Rate (Friday)	0.00%
Weekday No-Show Rate	11.41%
Weekend No-Show Rate	5.49%
ANOVA F-Statistic (Day)	6.4836
ANOVA p-value (Day)	< 0.001
Weekday vs. Weekend T-Test p-value	< 0.001
ANOVA F-Statistic (Time)	0.4196
ANOVA p-value (Time)	0.6574
Decision ($\alpha = 0.05$)	Reject H_0

Research Question 3: How do external environmental determinants, specifically government-declared holidays and PAGASA typhoon signal warnings, impact the statistical likelihood of patient cancellations?

H_0 : Government-declared holidays and local typhoon signals have no effect on patient attendance.

H_a : Patient cancellations are significantly higher on government-declared holidays and during local typhoon signals compared to normal days.

TABLE 9. HYPOTHESIS TESTING RESULT OF ENVIRONMENTAL FACTORS

Metric	Value
No-Show Rate (Typhoon)	40.77%
No-Show Rate (No Typhoon)	4.87%
Typhoon T-Statistic	15.5571
Typhoon p-value	< 0.001
No-Show Rate (Holiday)	10.47%
No-Show Rate (Regular Day)	7.82%
Holiday T-Statistic	0.8803
Holiday p-value	0.3789
Decision ($\alpha = 0.05$)	Reject H_0

The analysis leads to the rejection of the null hypothesis, identifying environmental conditions as the single most powerful determinant of patient attendance in this study. While government-declared holidays showed a slightly elevated non-attendance rate (10.47%) compared to regular days (7.82%), this difference was not statistically significant ($p=0.379$), suggesting that social calendar events are not a primary driver of cancellations. In contrary, the presence of a typhoon signal proved to be the single most disruptive factor in the entire dataset. The no-show rate during typhoons surged to 40.77%, nearly ten times the rate of clear days (4.87%), with an extremely high T-statistic (15.56) and a p-value of zero (< 0.001). This confirms that while social interruptions are manageable, severe physical barriers like weather events make attendance statistically impossible for a large segment of the patient population.

Research Question 4: Do significant differences in cancellation rates exist across different patient profiles defined by age groups, clinical diagnoses, and residential distance from the clinic?

H_0 : Cancellation rates do not vary across age groups, diagnosis categories, or residential distances.

H_a : Cancellation rates differ significantly among age groups, diagnosis categories, and residential distance categories.

TABLE 10. HYPOTHESIS TESTING RESULT OF DEMOGRAPHICS AND CLINIC RISK PROFILES

Metric	Value
Highest Diagnosis Rate (GDD)	12.82%
Highest Age Group Rate (7-9 years)	9.70%
Lowest Age Group Rate (3-4 years)	6.76%
Distance Rate (Moderate)	8.60%
Distance Rate (Far)	7.69%
ANOVA p-value (Age)	0.1903
ANOVA p-value (Diagnosis)	0.8095
ANOVA p-value (Distance)	0.7759
Decision ($\alpha = 0.05$)	Fail to Reject H_0

Contrary to the trends observed in the exploratory visualization, statistical testing indicates that demographic and clinical profiles are not significant predictors of non-attendance. The analysis fails to reject the null hypothesis

across all three categories. Although children aged 7-9 years exhibited a higher raw no-show rate (9.70%) compared to the 3-4-year-old group (6.76%), the ANOVA result ($p=0.190$) confirms this difference is not statistically significant. Similarly, while patients with Global Developmental Delay (GDD) appeared to have elevated cancellation rates (12.82%), the high p-value for diagnosis ($p=0.810$) suggests this variation is due to random chance rather than a systemic pattern. Furthermore, the "moderate distance paradox" observed earlier where those living at moderate distances seemed more likely to cancel, was proven statistically negligible ($p=0.776$). These results strongly suggest that who the patient is (age, diagnosis, location) matters far less than when the appointment is scheduled (day of week) or what external conditions exist (weather).

Research Question 5: Does a patient's historical attendance record, specifically the frequency of prior cancellations, serve as a significant predictor for future appointment non-attendance?

H_0 : A patient's previous cancellation behavior has no effect on their likelihood of cancelling future appointments.

H_a : Patients who cancelled previous appointments are significantly more likely to cancel subsequent appointments compared to those with consistent attendance.

TABLE 11. HYPOTHESIS TESTING RESULT OF HISTORICAL BEHAVIOR

Metric	Value
No-Show Rate (Has History)	10.44%
No-Show Rate (Clean History)	6.04%
Difference	+4.40%
T-Test Statistic	3.1396
T-Test p-value	0.0017
Correlation	0.0327
Decision ($\alpha = 0.05$)	Reject H_0

The statistical evidence supports the rejection of the null hypothesis, confirming that past behavior is a significant predictor of future attendance. Patients with a recorded history of non-attendance exhibited a no-show rate of 10.44%, which is nearly double the 6.04% rate observed among those with a clean attendance record. The T-test results ($t=3.14$, $p=0.002$) validate this difference as statistically significant. However, the low correlation coefficient ($r=0.033$) between the specific ratio of past no-shows and current behavior suggests a binary risk factor rather than a linear progression: the critical predictor is whether a patient has missed an appointment before, rather than how many they have missed in total. This implies that once a patient breaks the pattern of consistent attendance, their risk profile elevates permanently, regardless of the frequency of subsequent misses.

Predictive Model Performance

Table 12 presents comprehensive performance metrics for the three implemented models across the test set. The comparative evaluation of the three architectures reveals a clear trade-off between sensitivity and specificity. Logistic Regression demonstrated the highest Recall (0.708) and

ROC-AUC (0.777), making it the most effective model for identifying potential no-shows; however, its low Precision (0.230) indicates a substantial number of false positives, which could result in unnecessary administrative follow-ups.

TABLE 12. MODEL PERFORMANCE METRICS

Metric	Logistic Regression	Random Forest	Gradient Boosting
Accuracy	0.7881	0.9371	0.9338
Precision	0.2297	1.0000	0.5909
Recall	0.7083	0.2083	0.5417
F1-Score	0.3469	0.3448	0.5652
ROC-AUC	0.7774	0.7633	0.7214

In contrast, the Random Forest model exhibited perfect Precision (1.000) and the highest overall Accuracy (0.937), reflecting highly reliable predictions when a no-show is flagged. Nevertheless, its low Recall (0.208) indicates that it fails to detect most actual no-shows, limiting its usefulness for proactive intervention. Gradient Boosting provided the most balanced performance, achieving the highest F1-Score (0.565) while maintaining moderate Precision (0.591) and Recall (0.542). This balance enables it to identify more than half of no-show cases while controlling false alarms at a manageable level. Consequently, although Logistic Regression excels in maximizing risk detection, Gradient Boosting is selected as the optimal model due to its superior equilibrium between Precision and Recall, as reflected in its F1-Score.

TABLE 13. TOP FEATURES OF LOGISTIC REGRESSION

Rank	Feature	Coefficient
1	typhoon_signal_level	0.6158
2	log_duration_minutes	0.3954
3	lead_time_is_missing	0.3149
4	age	0.2484
5	has_down_payment_encoded	0.2301

TABLE 14. TOP FEATURES OF RANDOM FOREST

Rank	Feature	Importance
1	log_previous_appointment_count	0.1244
2	typhoon_signal_level	0.1208
3	noshow_ratio	0.0780
4	diagnosis_encoded	0.0694
5	log_lead_time_days	0.0660

TABLE 15. TOP FEATURES OF GRADIENT BOOSTING

Rank	Feature	Importance
1	typhoon_signal_level	0.2201
2	log_previous_appointment_count	0.1611
3	reminder_time_lag	0.0870
4	noshow_ratio	0.0710
5	log_lead_time_days	0.0579

Typhoon signal level emerged as a consistently strong predictor, ranking first in both Logistic Regression and Gradient Boosting and second in Random Forest, indicating that weather severity significantly influences attendance behavior. Historical engagement variables, such as previous appointment count and no-show ratio, were highly influential in tree-based models, suggesting that past attendance behavior is a strong determinant of future outcomes. Logistic Regression placed greater weight on session-specific and demographic factors (e.g., duration, age, and down payment status), while ensemble models emphasized behavioral history and timing-related features. Overall, the results

indicate that environmental risk and prior attendance patterns are the most stable predictors across models.

V. DISCUSSION

External environmental factors and temporal constraints are more significant determinants of attendance than a patient's specific clinical or demographic profile. Most notably, typhoon signal levels emerged as the strongest predictor of no-shows, with a positive correlation of $r = 0.375$ and a dramatic increase in missed appointments from 4.87% during clear weather to 40.77% during typhoons, highlighting the substantial physical barrier posed by severe weather in the Isabela region. Temporal factors also played a key role. Patients were roughly twice as likely to miss weekday appointments compared to weekend sessions, with weekend slots, comprising 58.1% of total appointments, exhibiting consistently low no-show rates (4-6%), whereas mid-week absences peaked on Wednesdays at 17.03%, reflecting how work schedules, school commitments, and logistical fatigue create "mid-week friction." Additionally, a recency effect was observed, where the outcome of the immediately preceding appointment ($r = 0.141$) predicted attendance more strongly than a patient's long-term history ($r = 0.067$), suggesting that short-term momentum influences behavior. Interestingly, while overall arrears status was a significant predictor ($p < 0.001$), session-specific down payments were not ($p = 0.435$), indicating that a patient's long-term financial relationship with the clinic better reflects reliability than a single transactional deposit.

The findings of this study both support and extend existing healthcare analytics literature. In terms of environmental integration, the results align with Liu et al., who observed that adverse weather increases no-show probabilities. However, this study adds a unique perspective by quantifying the effect of PAGASA typhoon signal levels specifically in the Philippine context. In contrast to prior work on demographic predictors, while Dunstan et al. identified residential distance as a primary factor in public healthcare settings, this study found distance to be statistically negligible ($p = 0.776$) in a private clinic, suggesting that patients in private practice may be more willing to travel long distances if weather and temporal conditions are favorable. Finally, the temporal dynamics observed, particularly the preference for weekend appointments, corroborate Hathaway et al.'s findings on day-of-week and time-of-day effects, yet the pronounced weekend advantage here appears to reflect local working patterns of Filipino parents, highlighting a distinct contextual characteristic.

This study has several limitations. First, the data were sourced from a single clinic in Isabela, so observed patterns may differ in urban centers like Metro Manila, where traffic congestion could replace weather as the primary environmental barrier. Second, the low baseline no-show rate of 7.97% resulted in a significant class imbalance, which constrained the recall of tree-based models such as Random Forest (0.208), making them overly conservative in predicting missed appointments. Finally, the dataset spans only February to October 2025, excluding the full annual

cycle and potentially missing attendance patterns related to the Christmas season or the start of the academic year.

Given the 40% non-attendance risk during typhoons, clinics should implement a "Typhoon Protocol," such as proactively adjusting schedules when Signal Level 1 or higher is declared. Since this clinic pre-assigns time slots well in advance, offering makeup classes may be logistically challenging. Instead, clinics could consider strategies such as temporarily overbooking mid-week sessions that coincide with forecasted typhoon days or creating a small buffer of flexible slots in advance to accommodate patients who miss appointments due to severe weather. With Wednesdays showing the highest no-show risk, clinics could implement higher-frequency or more personalized reminders for mid-week appointments to reduce "mid-week friction." Future research should incorporate additional socioeconomic indicators, such as a patient's primary mode of transportation (private car vs. public tricycle/jeepney), to better understand how weather, distance, and transport type interact in affecting attendance. Lastly, for practical clinical application, the Gradient Boosting model is recommended due to its superior F1-Score (0.565), offering a balanced approach between identifying no-shows and minimizing false alarms.

VI. CONCLUSION

This study developed a machine learning framework to predict appointment no-shows in a Philippine pediatric therapy clinic, achieving strong predictive performance through the integration of clinical administrative data and external environmental variables. The findings highlight that attendance behavior is shaped more by contextual and temporal constraints than by purely demographic or clinical characteristics.

The most significant predictor of non-attendance was tropical cyclone severity. No-show rates increased nearly tenfold during active typhoon signals, demonstrating that severe weather acts as a structural barrier to access rather than a minor inconvenience. Behavioral recency also emerged as a key factor, with a patient's most recent attendance outcome proving more predictive than their long-term history, suggesting that short-term behavioral momentum strongly influences future adherence. Temporal patterns revealed pronounced mid-week friction, while weekend appointments showed the highest stability, reflecting the impact of work and school schedules on family logistics. Financial analysis further indicated that overall arrears status was a meaningful predictor of reliability, whereas session-specific down payments were not, emphasizing the importance of long-term financial engagement over one-time deposits.

Among the evaluated models, Gradient Boosting is recommended for deployment due to its balanced performance (F1-Score = 0.565), offering a practical trade-off between identifying at-risk appointments and minimizing false alarms. Overall, this framework enables clinics to shift from reactive scheduling toward proactive, data-informed decision-making, improving resource allocation and continuity of care in weather-vulnerable regions.

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